

Problem-1

$$n = 20$$

$$p = 0.2$$

$$X \sim (n=20, p=0.2) \leftarrow \text{Binomial.}$$

Problem-2

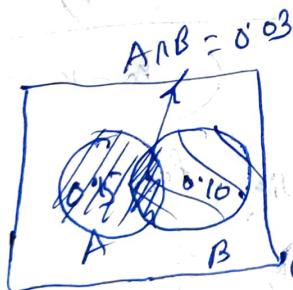
$$\text{Total possibility} = 62^8$$

$$62 = 26 + 26 + 10$$

Possibility of Password with No numeric = 52^8
So, Probability of Password with no digit = $\frac{52^8}{62^8}$
So, Probability of Password with at least one digit, $1 - \frac{52^8}{62^8}$

$$= 1 - \left(\frac{52}{62}\right)^8 = 1 - 0.29 = \boxed{0.76}$$

Problem-3



$$(a) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.03}{0.10} = 0.3$$

$$(b) P(A) = 0.15 \quad P(B) = 0.10$$

$$P(A^c) = 1 - 0.15 = 0.85 \quad (\text{Module A does not have bug})$$

$$\begin{aligned} P(B|A^c) &= \frac{P(B \cap A^c)}{P(A^c)} = \frac{P(B) - P(B \cap A)}{P(A^c)} \\ &= \frac{0.10 - 0.03}{0.85} \\ &= 0.07 / 0.85 \\ &= \boxed{0.08} \end{aligned}$$

Problem - 4

$$P(O_{\text{correct}} | P_{\text{correct}}) = 0.95$$

D^+ → Detect the object

$$P(D^+ | O_{\text{present}}) = 0.95 \quad P(D^+ | O_{\text{present}}^c) = 0.02$$

$$P(O_{\text{present}}) = 0.1$$

$$P(O_{\text{present}} | D^+)$$

$$= \frac{P(D^+ | O_{\text{present}}) \cdot P(O_{\text{present}})}{P(D^+ | O_{\text{present}}) \cdot P(O_{\text{present}}) + P(D^+ | O_{\text{present}}^c) \cdot P(O_{\text{present}}^c)}$$

$$= \frac{0.95 \times 0.1}{(0.95 \times 0.1) + (0.02 \times 0.99)}$$

$$= \frac{0.095}{0.095 + 0.0198} = 0.85$$

The obstacle is truly present, given the sensor detects as

$$\text{Obstacle} = 0.85$$

Problem - 5

$$P(E_{\text{span}}) = 0.1$$

$$P(\text{Money} | E_{\text{span}}) = 0.8$$

$$P(\text{Money} | E_{\text{span}}^c) = 0.05$$

$$P(E_{\text{span}}^c) = 0.99$$

$$P(E_{\text{span}} | \text{Money})$$

$$= \frac{P(\text{Money} | E_{\text{span}}) \cdot P(E_{\text{span}})}{P(\text{Money} | E_{\text{span}}) \cdot P(E_{\text{span}}) + P(\text{Money} | E_{\text{span}}^c) \cdot P(E_{\text{span}}^c)}$$

$$= \frac{0.8 \times 0.1}{(0.8 \times 0.1) + (0.05 \times 0.99)}$$

$$= \frac{0.08}{0.08 + 0.0495} = 0.61$$

$$= 0.61$$

28/29/21

Problem-6

I = Test

Positive

$$P(A_{\text{diseased}}) = 0.0001 = P(A_{\text{effected}}) = 1 - 0.0001 = 0.9999$$

$$P(I | D_{\text{present}}) = 0.99$$

$$P(I | D_{\text{absent}}) = 0.005$$

$$P(D_{\text{present}} | I) = \frac{0.99 \times 0.0001}{0.99 \times 0.0001 + 0.005 \times 0.9999} = \frac{0.000099}{0.000099 + 0.0049995} = \frac{0.000099}{0.0050985} = 0.01654$$

That means $\rightarrow 1.654\%$ chance he have the disease.

Problem-7

This is the problem of Binomial distribution

$$P(X=k) = {}^n C_k (p)^k (1-p)^{n-k}$$

$$n=20$$

$$k=3$$

$$(a) P(X=3) = {}^{20} C_3 (0.02)^3 (1-0.02)^{17}$$

$$P(\text{Defective}) = 0.02$$

$$= 0.0064$$

(b) At least 2 microscope are defective, that means

$$X=0 \leftarrow 0 \text{ defective}$$

$$X=1 \leftarrow 1 \text{ defective}$$

$$P(X=0) = {}^{20} C_0 (0.02)^0 (1-0.02)^{20} \approx 0.667$$

$$P(X=1) = {}^{20} C_1 (0.02)^1 (1-0.02)^{19} = 0.27$$

$$P(X \geq 2) = 1 - [P(X=0) + P(X=1)] = 0.060$$

$\rightarrow$ That means 6% will defective

Problem-8.

$$PDF = f(x) = \lambda e^{-\lambda x}$$

$$Mean = \frac{1}{\lambda} = 500 \Rightarrow \lambda = 1/500$$

$$Variance = 1/\lambda^2$$

x is the time that the m/c last between 400 to 600.

$$\begin{aligned} (a) \quad P(400 < x < 600) &= P(x > 400) - P(x > 600) \\ &= \lambda e^{-\lambda \times 400} - \lambda e^{-\lambda \times 600} \\ &= \lambda \left(e^{-\frac{400}{500}} - e^{-\frac{600}{500}} \right) \\ &= \lambda \left(e^{-4/5} - e^{-6/5} \right) \\ &= \frac{e^{-4/5} - e^{-6/5}}{500} = \frac{e^{-0.8} - e^{-1.2}}{500} \\ &= \frac{0.44 - 0.30}{500} = \frac{0.14}{500} \\ &= 0.00028 \end{aligned}$$

It is already

run 300 hours, ~~but~~ what is the probab. it run another 250 hours. $\rightarrow$ That means it should run $300 + 250 = 550$ hours

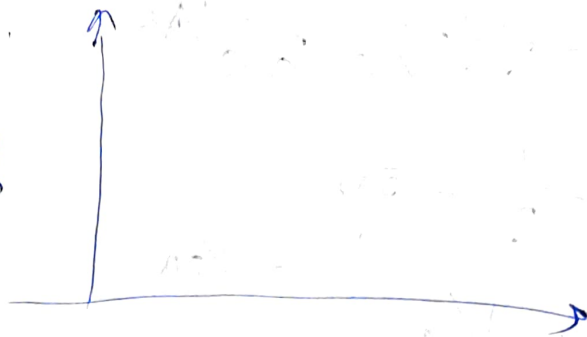
$$P(x > 550 | x > 300) \rightarrow$$

$$\text{Here } P(x > 300 + 250 | x > 300) = P(x > 250) \quad \uparrow \text{Memoryless Property}$$

$$P(x > 250) = \lambda e^{-\lambda x} = \frac{1}{500} \times e^{-250/500} = \frac{e^{-1/2}}{500} = 0.0012$$

P-9

Q

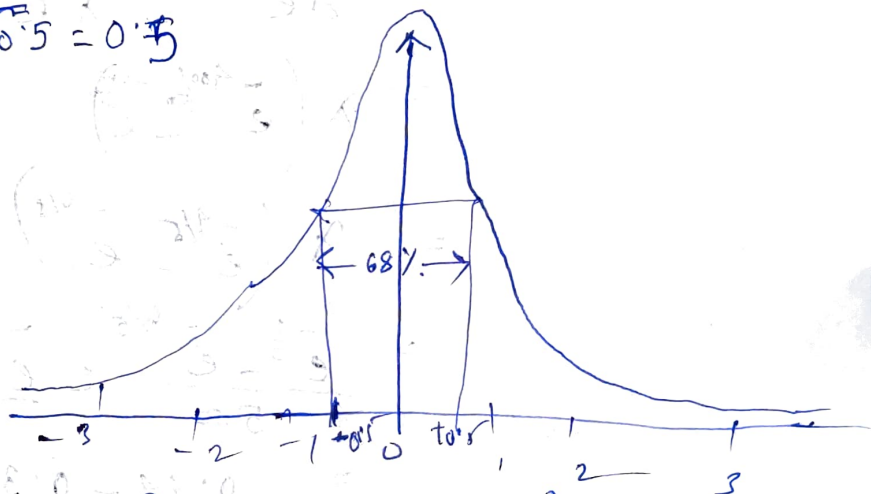


$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$= \frac{1}{0.5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x}{0.5}\right)^2}$$

$$\mu = 0$$

$$S.D. = \sigma = \sqrt{0.5} = 0.5$$



$$P(x > 0.3) - P(x > 0.2)$$

P-15.

Calculate the Eigen Value & Eigen Vector

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}$$

for A vector

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 5-\lambda & 2 \\ 2 & 5-\lambda \end{vmatrix} = (5-\lambda)(2-\lambda) - 4$$
$$= \lambda^2 - 7\lambda + 6 = 0$$
$$= (\lambda - 6)(\lambda - 1) = 0$$

$$\lambda = 6$$

$$\lambda_1 = 6, \lambda_2 = 1$$

$$(A - 6I) =$$

$$A\vec{x} = \lambda\vec{x}$$

$$\lambda = 6, \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 6 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$5x_1 + 2x_2 = 6x_1$$

$$2x_2 = x_1$$

$$\text{or, } x_2 = \frac{x_1}{2} \quad \text{--- (1)}$$

$$5x_1 + 2x_2 = 6x_1 \quad \text{--- (1)}$$

$$2x_1 + 2x_2 = 6x_2$$

$$2x_1 = 4x_2$$

$$1, \& \text{ (2) are same Eq. } \frac{x_1}{2} = x_2 \quad \text{--- (2)}$$

$$x_2 = 2, x_1 = 1$$

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is one eigen vector.

$$5x_1 + 2x_2 = x_1 \text{ or, } 4x_1 = 2x_2 \text{ or, } \underline{2x_1 = x_2}$$

$$2x_1 + 2x_2 = x_2 \text{ or, } 2x_1 = -x_2 \text{ or, } 2x_1 = -x_2$$

$$\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix} \quad \text{or, } (B - \lambda I) = 0$$

$$\det(B - \lambda I) = 0 \quad B - \lambda I = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 3 \\ -1 & 1 & 3 \end{bmatrix}$$

$$\det(B - \lambda I) = 0$$

$$(3 - \lambda) [(1 - \lambda)(2 - \lambda) - 1] = 0$$

$$\text{or, } \lambda = 3, 2, 1$$

$$\lambda = 1 \quad (B - 1I) = 0$$

$$(B - \lambda I) \vec{x} = \lambda \vec{x}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\text{or, } x_1 + x_2 = x_1 \quad \text{or, } x_2 = 0$$

$$2x_2 = x_2 \quad \text{or}$$

$$x_1 + x_2 + 3x_3 = x_3 \quad \text{or, } x_1 + x_2 = -3x_3$$

$$\lambda = 1 \quad \vec{x} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$x_1 = -3x_3$$

$$\lambda = 2 \quad \vec{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda = 3, \vec{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$